F70LA LIFE INSURANCE MATHEMATICS - Solutions to 2022/23 December Exam

1.
- They would require more evidence for a term insurance.
- With an endowment insurance they are sure to pay a benefit, but not for a term insurance.
- The loss on death near the end of the term is much smaller for an endowment insurance.
- For a term insurance, the premiums are lower in relation to the benefit paid out.

2. Yes, for $S_0(t) = 1 - \frac{t}{50}$, as it satisfies the three conditions:
- $S_0(0) = 1$.
- For a limiting age of $\omega = 50$ years, $\lim_{t\to\omega} S_0(t) = 1 - 50/50 = 0$.
- $dS_0(t)/dt = -1/50 < 0$.

3. First, A_{60} calculated at 12.36% is the same as ${}^2A_{60}$ calculated at 6%. .
From the Tables, ${}^2A_{60} = 0.14098$ calculated at 6% .
Then at the effective annual interest rate of 12.36%,

$$\begin{aligned}\ddot{a}_{60} =& \frac{1 - A_{60}}{d}\\ =& \frac{1 - 0.14098}{0.1236(1.1236)^{-1}}\\ =& 7.809.\end{aligned}$$

4. (a) 1 mark for each point to a maximum of 2 marks.
- q_x increases from 0.2 at age 90 to 0.4 at age 100.
- This is very similar to ELT15 for males, in which $q_{90} = 0.20465$ and $q_{100} = 0.38705$. This suggests that it is possible.
- However, the increase is linear rather than exponential, which gives very different intermediate values from Makeham's Law or Gompertz's Law (which are both quite realistic at these high ages).

(b) (i)
$$_3p_{95} = p_{95}p_{96}p_{97} = 0.7 \times 0.68 \times 0.66 = 0.31416.$$

(ii) As
$$_3p_{91} = p_{91}p_{92}p_{93} = 0.78 \times 0.76 \times 0.74 = 0.438672$$

and

$$ {}_2q_{94} = 1 - p_{94}p_{95} = 1 - 0.72 \times 0.70 = 0.496, $$

$$ {}_3|_2q_{91} = {}_3p_{91}\,{}_2q_{94} = 0.438672 \times 0.496 = 0.21758. $$

(c) As $q_{93} = 0.26$, $q_{94} = 0.28$ and, from part(b)(i), ${}_3p_{95} = 0.31416$,

$$ \begin{aligned} {}_5p_{[93]} &= p_{[93]}\,p_{[93]+1}\,{}_3p_{95} \\ &= (1 - 2q_{93})(1 - 1.5q_{94})\,{}_3p_{95} \\ &= 0.48 \times 0.58 \times 0.31416 = 0.08746. \end{aligned} $$

(d) 1 mark for each point to a maximum of 2 marks.

- The select mortality is worse than the ultimate mortality, e.g. it is twice as high over the first year after selection.
- However, two years after selection, the selection effect has worn off and the mortality is the same as under the ultimate table.

5. (a) (i) First calculate

$$ A^{(2)}_{\substack{1 \\ 74:\overline{1}|}} = v^{0.5}\,{}_{0.5}q_{74} + v\,{}_{0.5}p_{74}\,{}_{0.5}q_{74.5}. $$

Under UDD, ${}_{0.5}q_{74} = 0.5q_{74}$ and

$$ {}_{0.5}q_{74.5} = \frac{0.5q_{74}}{1 - {}_{0.5}q_{74}} = \frac{0.5q_{74}}{{}_{0.5}p_{74}}. $$

Hence with $v = 1/1.09$,

$$ A^{(2)\,1}_{74:\overline{1}|} = (1.09)^{-0.5}(0.5q_{74}) + (1.09)^{-1}(0.5q_{74}) = 0.03904286 $$

and the premium is $500\,000A^{(2)\,1}_{74:\overline{1}|} = \$19\,521.43$.

(ii) Under a constant force of mortality,

$$ \mu = -\ln(1 - 0.04164) = 0.042532. $$

As

$$ \mathbb{P}[\text{death in first half of year}] = 1 - \exp(-0.042532 \times 0.5) = 0.02104147 $$

and either

$$ \begin{aligned} \mathbb{P}[\text{death in second half of year}] &= \exp(-0.042532 \times 0.5) - \exp(-0.042532 \times 1) \\ &= 0.02059863, \end{aligned} $$

or $(1 - 0.02104137) \times 0.02104137 = 0.02059863$, we get

$$ \begin{aligned} \text{Premium} &= 500\,000(0.02104137 \times v^{0.5} + 0.02059863 \times v) \\ &= 10\,076.99 + 9\,448.91 = \$19\,525.90. \end{aligned} $$

(b) 1 mark for each point to a maximum of 3 marks.

- The answers are similar, as expected.
- Probability of dying is larger under constant force of mortality, compared to UDD.
- The difference is small enough to be unimportant, considering the difficulty in choosing the numbers for the premium basis.
- Constant μ gives a higher probability of dying in the first half of the year...
- ...and hence a higher premium,...
- ...because of the discounted value of money.
- Both methods give the same total probability of a payment being made $= q_{74}$.
- If the interest rate were lower (or zero) then the difference would be smaller (or zero).

(c) 1 mark for each point to a maximum of 2 marks.

- The premium would be higher,
- Because the benefit would be paid sooner.

6. The profit random variable is

$$P = 11\,700 - 1\,000 a_{\overline{K_{60}|}},$$

where the random variable K_{60} is the future curtate lifetime in years from age 60.

(a) The expected value of P is

$$11\,700 - 1\,000 a_{60} \ = 11\,700 - 1\,000 \times 10.891 = \text{£}809.$$

(b) As $A_{60} = \mathbb{E}\left(v^{K_{60}+1}\right)$ and $d = iv$,

$$\begin{aligned}\operatorname{var}\left(a_{\overline{K_{60}|}}\right) = \operatorname{var}\left(\frac{1-v^{K_{60}}}{i}\right) = \frac{1}{(iv)^2}\operatorname{var}\left(v^{K_{60}+1}\right) &= \frac{1}{d^2}\left({}^2A_{60} - A_{60}^2\right) \\ &= \frac{0.14098 - (0.32692)^2}{(1-1.06^{-1})^2} \\ &= 10.644.\end{aligned}$$

Then the standard deviation of P is

$$\mathrm{SD}(P) = \mathrm{SD}(1\,000\, a_{\overline{K_{60}|}}) = 1\,000\sqrt{10.644} = \text{£}3\,263.$$

(c) As $P = 1\,000(11.7 - a_{\overline{K_{60}|}})$,

$$\mathbb{P}[P > 0] = \mathbb{P}\left[11.7 > a_{\overline{K_{60}|}}\right] = \mathbb{P}\left[v^{K_{60}} > 1 - 11.7i\right].$$

Taking natural logs of both sides,

$$\begin{aligned}\mathbb{P}[P > 0] = \mathbb{P}\left[K_{60} < -\frac{\ln(1-11.7i)}{\ln(1+i)}\right] &= \mathbb{P}[K_{60} < 20.8] \\ &= \mathbb{P}[K_{60} \leq 20] \\ &= \mathbb{P}[K_{60} + 1 \leq 21] \\ &= {}_{21}q_{60}.\end{aligned}$$

So

$$\mathbb{P}[P > 0] = {}_{21}q_{60} = 1 - {}_{21}p_{60} = 1 - \frac{\ell_{81}}{\ell_{60}} = 1 - \frac{4\,901.4789}{9\,287.2164} = 1 - 0.52777 = 0.47223.$$

and the probability that the life office's profit is positive is 0.47223.

(d) Maximum 2 marks.

- As calculated in part(c), there is a 53% chance of a loss.
- The maximum possible loss is £$1000/i$ = £16 666.67 which gives a loss of £4 966.67.
- The maximum possible profit is £11 700, corresponding to the person dying before age 61.
- Therefore, the present value of the profit lies in the interval $(-4\,966.67, 11\,700]$. The loss can be large, about 44% of the price of the annuity (11 400/4 966.67).
- The expected present value of the profit is positive at £809. However, the standard deviation is more than four times as large, at £3 263. This suggests that the distribution of the present value of the profit is spread out.

(e) It would not change at all. The only difference would be a payment at age 60 with probability 1, which would not affect the variance (or standard deviation).

7. (a)

$$\mathbb{E}\text{PV}(\text{Premiums}) = P\ddot{a}^{(4)}_{30:\overline{35}|}.$$

From the Tables, page 36, we have the approximation

$$\ddot{a}^{(4)}_{30:\overline{35}|} \approx \ddot{a}_{30:\overline{35}|} - \frac{3}{8}\left(1 - v^{35}\,{}_{35}p_{30}\right).$$

From the Tables, $\ddot{a}_{30} = 21.834$, $\ddot{a}_{65} = 12.276$ and so

$$\begin{aligned}\ddot{a}_{30:\overline{35|}} =& \ddot{a}_{30} - v^{35}\,{}_{35}p_{30}\,\ddot{a}_{65}\\ =& 21.834 - (1.04)^{-35}\,\frac{8\,821.2612}{9\,925.2094}\,(12.276)\\ =& 21.834 - 2.765 = 19.069.\end{aligned}$$

Hence

$$\ddot{a}^{(4)}_{30:\overline{35|}} \approx 19.069 - \frac{3}{8}\,(1 - 0.225229) = 18.778.$$

and $\mathbb{E}\text{PV}(\text{Premiums}) = 18.778P$.

(b)

$$\mathbb{E}\text{PV}(\text{benefits}) = 400\,000\bar{A}_{30} + 0.025(400\,000)(IA)_{30}.$$

Under AM92 at an interest rate of 4% per annum effective, from the Tables we find

$$A_{30} = 0.16023 \qquad \text{and} \qquad (IA)_{30} = 6.91559.$$

- Either: Under claims acceleration, $\bar{A}_{30} \approx (1.04)^{0.5}(0.16023) = 0.16340$ and $(I\bar{A})_{30} \approx (1.04)^{0.5}(6.91559) = 7.05255$. Hence

  $\mathbb{E}\text{PV}(\text{benefits}) = 400\,000(0.16340)+10\,000(7.05255) = \text{£}135\,885.50.$(Cl.Acc.)

- Or (same mark distribution): Under UDD, $\bar{A}_{30} \approx \frac{0.04}{\ln(1.04)}(0.16023) = 0.16341$ and $(I\bar{A})_{30} \approx \frac{0.04}{\ln(1.04)}(6.91559) = 7.05300$. Hence

  $\mathbb{E}\text{PV}(\text{benefits}) \approx 400\,000(0.16341)+10\,000(7.05300) = \text{£}135\,894.00.$ (UDD)

(c) Show that the expected present value of the expenses can be written as $aP + b$, where a and b are constants to be calculated.

$$\mathbb{E}\text{PV}(\text{expenses}) = 300 + 0.4\frac{P}{4} + 20a_{30} + 200\bar{A}_{30}.$$

Under AM92 at an interest rate of 4% per annum effective, from the Tables we find

$$\ddot{a}_{30} = 21.834 \qquad \Rightarrow \qquad a_{30} = \ddot{a}_{30} - 1 = 20.834.$$

From part(b), we have already calculated that

- Under claims acceleration, $\bar{A}_{30} \approx 0.16340$.
- Under UDD, $\bar{A}_{30} \approx 0.16341$.

Therefore

$$\begin{aligned}\mathbb{E}\text{PV}(\text{expenses}) =& 0.1P + 300 + 20a_{30} + 200\bar{A}_{30}\\ \approx& 0.1P + 300 + 20(20.834) + 200(0.16340)\\ \approx& 0.1P + 749.36.\end{aligned}$$

Therefore, $a = 0.1$ and $b = 749.36$.

(d) The annual premium P is the solution to

$\mathbb{E}\text{PV(premiums)} = \mathbb{E}\text{PV(benefits)} + \mathbb{E}\text{PV(expenses)} + \mathbb{E}\text{PV(profit)}$,

i.e. either: under claims acceleration,

$$18.778P = 135\,885.50 + (0.1P + 749.36) + 3\,000 \quad \Rightarrow \quad P = \frac{139\,634.86}{18.678} = \text{£}7\,475.90$$

or: under UDD,

$$18.778P = 135\,894.00 + (0.1P + 749.36) + 3\,000 \quad \Rightarrow \quad P = \frac{139\,643.36}{18.678} = \text{£}7\,476.36.$$